Chapter 3 Revision Questions

SDS 188

2023-03-15

Question 1

Given the price of a share L at 8 periods in time:

Time	1	2	3	4	5	6	7	8
L	1720	1880	2780	2040	2060	2860	2230	1370

(a) *Calculate the mean ($\bar{x}$) share price of L.*

$$\bar{x} = 2117.5$$

(b) *Calculate the geometric mean ($\bar{x}_G$) share price of L.*

$$\bar{x}_G = 2064.56$$

(c) *Calculate the geometric mean rate at which the investment grew each year ($\bar{r}_G$):*

	r_0	r_1	r_2	r_3	r_4	r_5	r_6	r_7
r_i	-	9.30%	47.87%	-26.62%	0.98%	38.83%	-22.03%	-38.57%

$$\bar{r}_G = -0.028063 \approx -2.8063\%$$

Question 2

Use the data from Question 1 to answer the following:

(a) *Determine the sample variance (s^2) of the price of share L:*

$$\begin{aligned}
s^2 &= \sum_{i=1}^{n} \frac{1}{n-1}(x_i - \frac{1}{n}\sum_{j=1}^{n} x_j)^2 \\
&= \frac{1}{n-1}\left[\sum_{i=1}^{n} x_i^2 - \frac{1}{n}\left(\sum_{j=1}^{n} x_j\right)^2\right] \\
&= \frac{1}{7}\left[37655800 - \frac{1}{8}\left(16940\right)^2\right] \\
&= \frac{1}{7}\left[37655800 - 35870450\right] \\
&= 255050
\end{aligned}$$

(b) *Calculate the standard deviation* (s):

$$
\begin{aligned}
s &= \sqrt{255050} \\
&= 505.0248
\end{aligned}
$$

(c) *Calculate the Coefficient of Variation of the price of L:*

$$
\begin{aligned}
CV_L &= \frac{505.0248}{2117.5} \\
&= 0.2385005 \\
&\approx 23.85\%
\end{aligned}
$$

Question 3

30 Different prices were collected for a single unit of commodity U. The prices are given in the form of a grouped data summary below:

Price Range	Frequency(f)	Midpoint(x)	$f \cdot x$	x^2	$f \cdot x^2$
1-20	2	10.5	21	110.25	220.5
21-40	6	30.5	183	930.25	5581.5
41-60	7	50.5	353.5	2550.25	17851.75
61-80	8	70.5	564	4970.25	39762
81-100	4	90.5	362	8190.25	32761
101-120	3	110.5	331.5	12210.25	36630.75
Total	**n=30**	-	**1815**	-	**132807.5**

(a) *Approximate the mean*

$$
\bar{x} = \frac{1815}{30} = 60.5
$$

(b) *Approximate the variance*

$$
\begin{aligned}
s^2 &= \sum_{i=1}^{n} \frac{1}{n-1}(x_i - \frac{1}{n}\sum_{j=1}^{n} x_j)^2 \\
&= \frac{1}{n-1}\left[\sum f \cdot x^2 - \frac{1}{n}\left(\sum f \cdot x\right)^2\right] && \text{Rewrite in alternative form} \\
&= \frac{1}{29}\left[132807.5 - \frac{1}{30}\left(1815\right)^2\right] \\
&= 793.1034
\end{aligned}
$$

Question 4

(a) *Given a RV X that follows a Normal distribution with a mean of 50 and a variance of 25, find the z-scores for the following three values:*

$$x_1 = 27.5 \qquad z_1 = -4.5$$

$$x_2 = 49 \qquad z_2 = -0.2$$

$$x_3 = 60 \qquad z_3 = 2$$

(b) *Which one of these observations can be considered an outlier? Motivate.*

x_1 can be considered an outlier as its z-score $>\mid 3 \mid$.

Question 5

You are given the following stem and leaf diagram:

1	3789
2	158999
3	0014
4	455
5	25
6	9

(a) *Find the range*

Maximum = 69
Minimum = 13
Range = $69 - 13 = 56$

(b) *Find the median*

Median position:

$$\frac{(n+1)}{2} = \frac{(20+1)}{2}$$

$$= 10.5$$

Taking the average of x_{10} and x_{11}:

$$\frac{(29+30)}{2}$$

$$= 29.5$$

(c) *Give the mode*

29

(d) *Comment on the shape of the distribution*

The distribution is skewed to the right.

Question 6 (Example 3.13 in the prescribed textbook)

Nutritional data about a sample of seven breakfast cereals includes the number of calories per serving. Compute the five-number summary of the number of calories in cereals:

Cereal	**Calories**
Kellogg's All Bran	80
Kellogg's Corn Flakes	100
Wheaties	100
Nature's Path Organic Multigrain Flakes	110
Kellogg's Rice Krispies	130
Post Shredded Wheat Vanilla Almond	190
Kellogg's Mini Wheats	200

Min = 80
Max = 200

Q1 position:

$$\frac{(n+1)}{4} = \frac{(7+1)}{4} = 2$$

Median/Q2 position:

$$\frac{(n+1)}{2} = \frac{(7+1)}{2} = 4$$

Q3 position:

$$\frac{3(n+1)}{4} = \frac{3(7+1)}{4} = 6$$

The five-number summary is:

Min	Q1	Q2	Q3	Max
80	100	110	190	200

Question 7

A population of 2-liter bottles of cola is known to have a mean fill-weight of 2.06 liters and a standard deviation of 0.02 liter.

(a) *The population is known to be bell-shaped. Describe the distribution of fill-weights. Is it very likely that a bottle will contain less than 2 liters of cola?*

$$\begin{aligned}\mu \pm \sigma &= 2.06 \pm 0.02 = (2.04, 2.08)\\ \mu \pm 2\sigma &= 2.06 \pm 2(0.02) = (2.02, 2.10)\\ \mu \pm 3\sigma &= 2.06 \pm 3(0.02) = (2.00, 2.12)\end{aligned}$$

Using the empirical rule, you can see that approximately 68% of the bottles will contain between 2.04 and 2.08 liters, approximately 95% will contain between 2.02 and 2.10 liters, and approximately 99.7% will contain between 2.00 and 2.12 liters. Therefore, it is highly unlikely that a bottle will contain less than 2 liters.

(b) *Suppose now that the shape of the population is unknown, and you cannot assume that it is bell-shaped. Describe the distribution of fill-weights. Is it very likely that a bottle will contain less than 2 liters of cola?*

$$\begin{aligned}\mu \pm \sigma &= 2.06 \pm 0.02 = (2.04, 2.08)\\ \mu \pm 2\sigma &= 2.06 \pm 2(0.02) = (2.02, 2.10)\\ \mu \pm 3\sigma &= 2.06 \pm 3(0.02) = (2.00, 2.12)\end{aligned}$$

Because the distribution may be skewed, you cannot use the empirical rule. Using Chebyshev's theorem, you cannot say anything about the percentage of bottles containing between 2.04 and 2.08 liters. You can state that at least 75% of the bottles will contain between 2.02 and 2.10 liters and at least 88.89% will contain between 2.00 and 2.12 liters. Therefore, between 0 and 11.11% of the bottles will contain less than 2 liters.

(c) *For some RV X that is not normally distributed, with mean 50 and variance of 25, find the bounds for the interval that should contain at least 95% of the data.*

Given Chebyshev's theorem:

$$P(\mu - k\sigma \leq x \leq \mu + k\sigma) \geq \left(1 - \frac{1}{k^2}\right)$$

$$P(50 - k \cdot 5 \leq x \leq 50 + k \cdot 5) \geq \left(1 - \frac{1}{k^2}\right)$$

Find the value for k (round final answer to 2 decimals):

$$
\begin{aligned}
\left(1 - \frac{1}{k^2}\right) &= 0.95 \\
-\frac{1}{k^2} &= -0.05 \\
k^2 &= \frac{1}{0.05} \\
k &= \sqrt{20} \\
k &= 4.4721 \\
k &\approx 4.47
\end{aligned}
$$

$$
\begin{aligned}
P(50 - 4.47 \cdot 5 \leq x \leq 50 + 4.47 \cdot 5) \geq 0.95 \\
P(27.65 \leq x \leq 72.35) \geq 0.95
\end{aligned}
$$

Thus, the interval containing atleast 95% of the data is $[27.65; 72.35]$

Question 8

A sample of size 10 is collected on two variables, X and Y:

X	16	18	18	19	21	14	18	26	17	21
Y	47	48	57	50	50	58	52	43	46	47

(a) *Calculate the covariance*

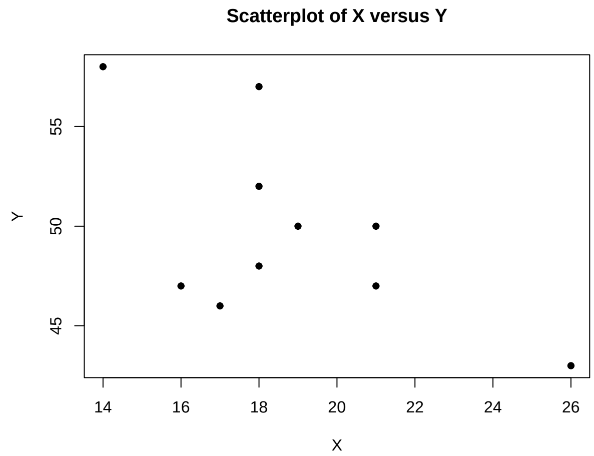
$$
\begin{aligned}
cov(X, Y) &= \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y}) \\
&= \frac{1}{n-1} \left(\sum_{i=1}^{n} x_i y_i - \frac{1}{n} \left[\sum_{i=1}^{n} x_i \sum_{i=1}^{n} y_i \right] \right) \\
&= -9.488889
\end{aligned}
$$

(b) *Give a disadvantage of the covariance*

It shows the direction of relationship, but cannot show the relative strength of relationship.

(c) *Calculate the correlation*

$$
\begin{aligned}
r &= \frac{cov(X, Y)}{s_x s_y} \\
&= \frac{-9.4889}{3.29309 \cdot 4.756282} \\
&= -0.6058207
\end{aligned}
$$


Scatterplot of X versus Y
Y
55
50
45
14
16
18
20
22
24
26
X